Pratik Padole

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Professional Summary

Artificial Intelligence and Data Science undergraduate with practical experience in Machine Learning, Deep Learning, Natural Language Processing, and Large Language Models. Proficient in Python, Scikit-learn, TensorFlow, and SQL, with expertise in developing end-to-end machine learning pipelines and AI-powered solutions.

Technical Skills

Languages: Python, SQL, Java, JavaScript
Machine Learning: Scikit-Learn, TensorFlow, PyTorch, XGBoost, Feature Engineering, Model Evaluation, Hyperparameter Tuning
Deep Learning & AI: CNNs, Computer Vision, Natural Language Processing (NLP), Generative AI, Large Language Models (LLMs), Prompt Engineering
Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn
Databases: MySQL, MongoDB
Cloud & Tools: AWS (Basics),REST APIs, Docker, Git, GitHub, Postman

Education

Dr. D. Y. Patil College of Engineering, Akurdi, Pune 2023 – 2027
Bachelor of Engineering in Artificial Intelligence and Data Science
CGPA: 9.35 / 10

Projects

TalentMatch AI GitHub
Python, FastAPI, Sentence Transformers, Scikit-Learn, Pandas, Docker

- Developed an AI-powered resume screening platform that ranks candidates against job descriptions using transformer-based semantic embeddings and cosine similarity.
- Built an end-to-end NLP pipeline for PDF parsing, text preprocessing, skill extraction, embedding generation, and weighted candidate ranking.
- Designed FastAPI-based REST APIs supporting multi-resume uploads, explainable match scores, and skill gap analysis.
- Optimized inference through embedding caching and batch processing, and containerized the application using Docker for scalable deployment.

PsyPredict: Mental Health Score Prediction using Machine Learning GitHub
Python, Scikit-Learn, Pandas, NumPy, Flask, Matplotlib

- Developed an end-to-end ML pipeline to predict students' mental health scores using lifestyle and behavioral features.
- Trained and compared Linear Regression, Decision Tree, Random Forest, and Gradient Boosting models with GridSearchCV and 5-fold cross-validation.
- Evaluated models using MAE, RMSE, and R^2 score to select the best-performing model.
- Deployed the optimized model as a Flask web application for real-time predictions and personalized recommendations.

Student Performance Prediction System GitHub
Python, Scikit-Learn, Pandas, NumPy, Matplotlib, Flask

- Developed an end-to-end machine learning pipeline to predict student mathematics performance using demographic and academic features.
- Performed data preprocessing, feature engineering, one-hot encoding, feature scaling, and exploratory data analysis.
- Trained and compared multiple regression models including Linear Regression, Decision Tree, Random Forest, and XGBoost using cross-validation.
- Evaluated model performance using MAE, RMSE, and R^2 score, and deployed the best-performing model through a Flask web application.

Relevant Coursework

Machine Learning, Artificial Intelligence, Computer Vision, Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Software Engineering

Achievements

- Solved 500+ Data Structures and Algorithms problems on LeetCode.
- Maintained a CGPA of 9.35/10 in Artificial Intelligence and Data Science.

Positions of Responsibility

Technical Lead – Society For Data Science (S4DS) Jul 2025 – May 2026

- Conduct technical sessions and mentor students in backend development and project building.

Web Lead – Intel AI Student Club (IASC) Aug 2025 – May 2026

- Lead web development initiatives for technical events, workshops, and student activities.